Vyas Sriman

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Education

University of Texas at Austin - Expected Graduation: May 2028
Bachelors of Science in Statistics & Data Science + Mathematics | GPA 4.0/4.0
Relevant Coursework: Machine Learning, Probability & Inference, Software Design, Linear Algebra, Data Structures & Algorithms

Experience

Software Engineering Intern, JP Morgan Chase **June 2026 - August 2026**

- Built a PyTorch sentiment model and led 3 engineers scaling its Java + FastAPI layer to 3+ teams on Kubernetes at p99 under 500ms
- Integrated the model into a Django + Bootstrap app delivering client sentiment predictions, cutting reporting time 50% for 4+ teams
- Caught the first production accuracy drop 6 weeks post-launch via nightly drift and vocabulary-shift tracking with one-click retraining

Undergraduate Researcher – EEG Foundation Models, University of Texas at Austin **May 2026 - September 2026**

- Improved brain-computer interface decoding accuracy by 4+ F1 points over prior baselines by running a 200-trial distributed Optuna hyperparameter sweep with TPE sampling and median pruning across 3 GPUs, fine-tuning a LoRA-adapted transformer in PyTorch
- Built a live brain-signal-to-screen-control system with sub-200ms response time using PyTorch, ONNX Runtime, and asyncio consuming OpenBCI data via LSL, with React + WebSocket frontend translating decoded motor imagery into real-time cursor control
- Cut experiment reproduction from days to under 10 minutes across 40+ checkpoints by scaffolding an MLflow registry with AI-assisted development and validation, auto-logging training configs, dataset versions, hyperparameters, and accuracy metrics per run

Undergraduate Researcher – Quantum Computing, University of Texas at Austin **January 2026 - May 2026**

- Cut quantum error-correction boilerplate from 50+ lines of custom code to 2 lines per experiment, and hardware credit usage by 30%, by shipping a pip-installable Python package with decorator-based ZNE and PEC APIs and SQLite-backed calibration caching
- Replaced a manual, error-prone quantum experiment workflow with an automated CI/CD pipeline using GitHub Actions and YAML-configured experiment definitions, auto-generating publication-ready LaTeX figures and catching 6 circuit bugs pre-submission
- Engineered a 50-node distributed network simulator with graph-based topology modeling in NetworkX and Monte Carlo statistical analysis over 1,000+ simulated trials, parallelized with Python multiprocessing and submitted as a peer-reviewed workshop paper

Projects

AI Resume Analyzer (2026) - Link

- Shipped a serverless resume analysis web app 40% faster with AI-assisted development under documented peer review, pairing React Router v7, TypeScript, and Tailwind CSS while orchestrating 4 Claude API agents for scoring, gap analysis, and bullet rewriting
- Increased repeat usage 40% by running client-side A/B tests instrumented with IndexedDB analytics, discovering that 70% of users ignored detailed feedback, then redesigning the UI around a quick-wins summary panel with interactive D3.js confidence visualizations

Touchline Value — Player Valuation Platform (2026) - Link

- Built a React and TypeScript valuation engine for 20 Premier League players, processing 100+ attributes across five weighted signals to generate real-time estimates and ±15% confidence ranges, with signal contribution breakdowns validated across 50+ scoring scenarios
- Engineered searchable rankings and a three-variable simulator supporting 600+ valuation combinations with sub-100 ms updates, configurable confidence ranges, and 85% unit-test coverage, reducing five-step manual comparisons to one responsive workflow

Podcast Knowledge Graph Builder (2026)

- Built an RSS ingestion pipeline with WhisperX ASR, pyannote diarization, and LLM entity/relation extraction across 500+ episodes
- Achieved p95 retrieval latency under 400ms by designing hybrid Neo4j graph and Qdrant vector search with Reciprocal Rank Fusion
- Reduced duplicate entities 60%+ via two-pass dedup using RapidFuzz string matching and embedding cosine similarity scoring
- Improved fault tolerance via Celery and Redis async task orchestration with idempotent SHA256 dedup and dead-letter queuing
- Delivered per-claim citations tied to speaker and timestamp by shipping a streaming RAG chat REST API over FastAPI and SSE
- Maintained answer quality across releases by building a CI/CD GitHub Actions eval harness with LLM-as-judge regression gating

Organizations

Texas Luminescence, Backend Engineer - Student software engineering org building and shipping full-stack applications
Texas BAXA, Member/Consultant - Business analytics association providing data-driven consulting to Austin-area clients

Skills

Languages & Databases: Python, SQL, Java, TypeScript, JavaScript, R, PostgreSQL, Neo4j, Qdrant, Redis, SQLite, Supabase, Prisma
Machine Learning & AI: PyTorch, Hugging Face Transformers, scikit-learn, PEFT, LoRA, ONNX Runtime, Optuna, MLflow, RAG, vector search, embeddings, WhisperX, ASR, NLP, LLM fine-tuning, knowledge graphs, model monitoring, transfer learning, MLOps
Backend & Frontend: FastAPI, Django, Flask, Celery, Node.js, Next.js, React, Tailwind CSS, REST APIs, WebSockets, asyncio, D3.js
Infrastructure & Tools: Docker, Kubernetes, Helm, AWS, GitHub Actions, Git, Unix/Linux, Grafana, Prometheus, pytest, Playwright
Data & Methods: Pandas, NumPy, NetworkX, distributed systems, microservices, CI/CD, data pipelines, A/B testing, observability, Agile